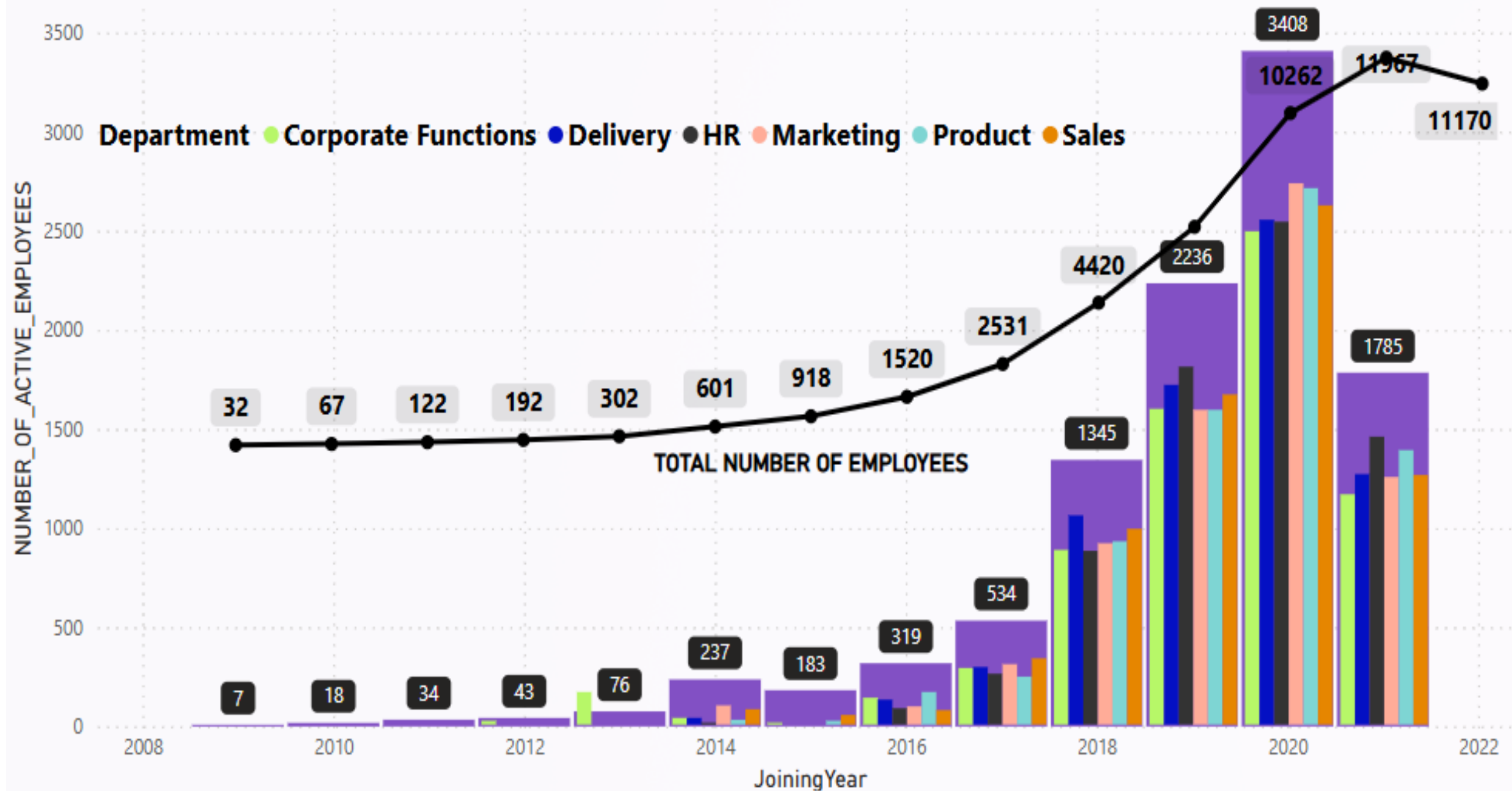


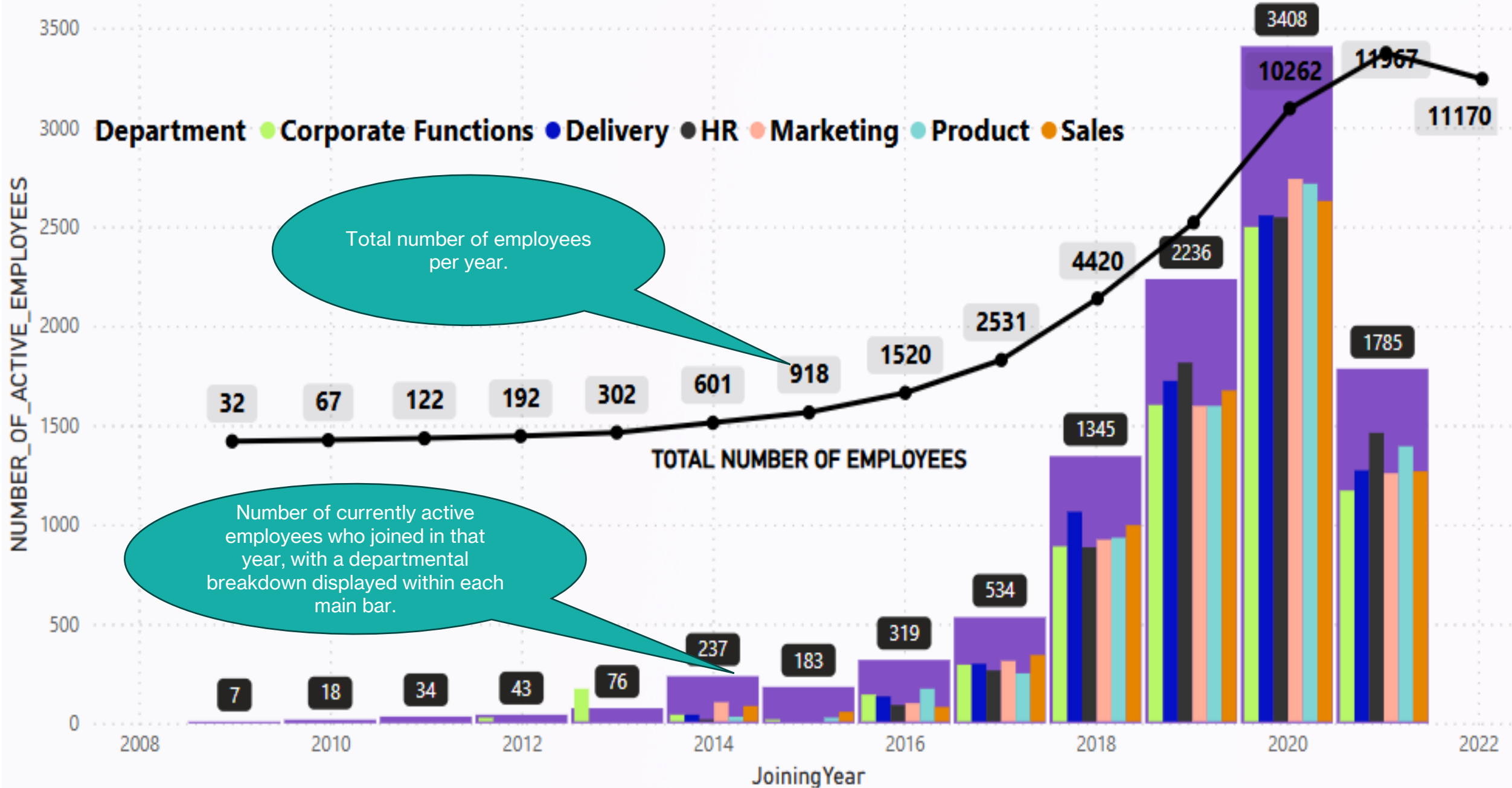


**NUMBER OF TOTAL
EMPLOYEES
THROUGHT YEARS
AND NUMBER OF
ACTIVE EMPLOYEES
BY YEAR OF JOINING
CHART VISUAL**

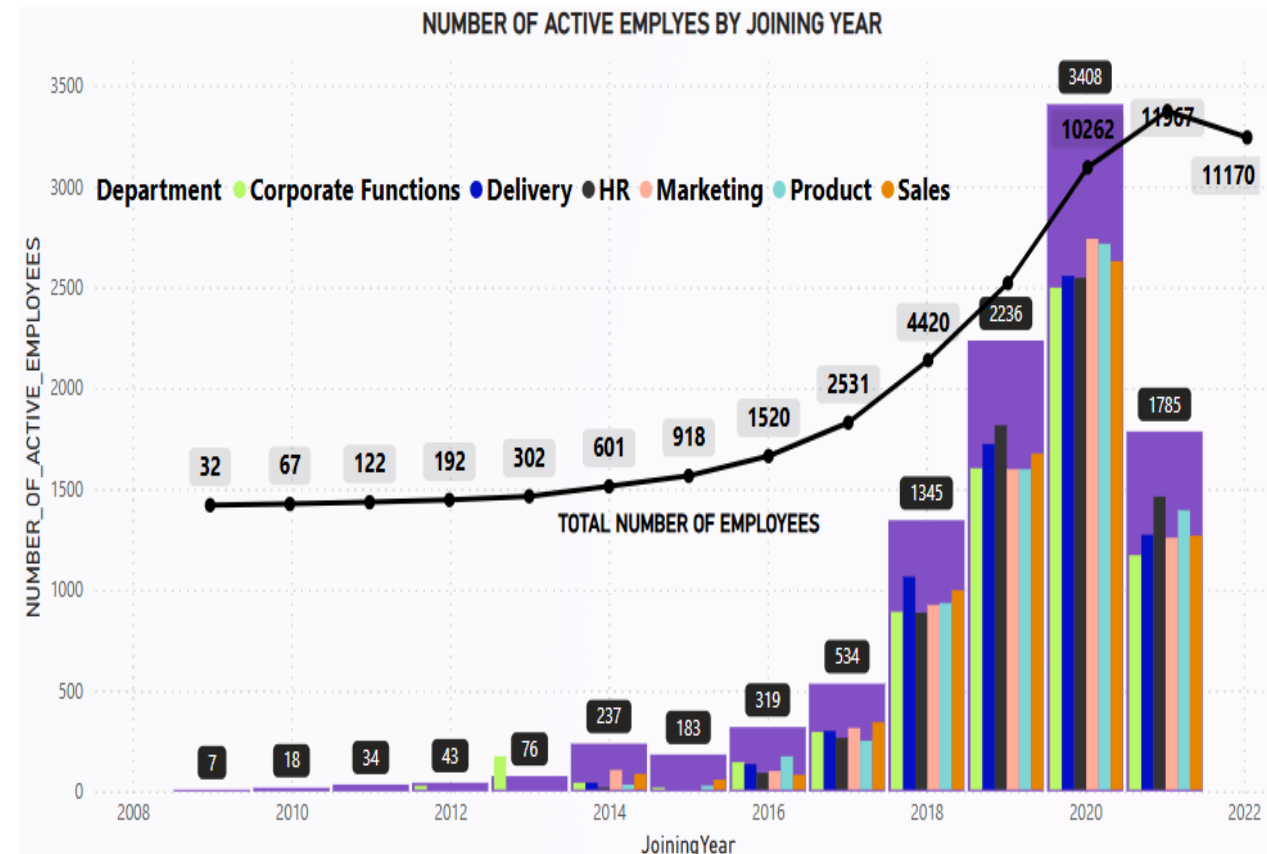
NUMBER OF ACTIVE EMPLYES BY JOINING YEAR



NUMBER OF ACTIVE EMPLOYEES BY JOINING YEAR



- The number of employees increased steadily from 2008 to 2021, followed by a slight decrease in 2022. In the current employee structure, most employees joined in 2020. This sharp rise in employee numbers follows a pattern typical of logistic regression.



Why Logistic Regression Is Often Used to Describe Employee Growth in Companies:

1. S-Shaped Growth Curve (Sigmoid Function)

- Logistic regression models growth using an **S-shaped (sigmoid) curve**.
- This curve matches how employee numbers typically grow:
 - Initial phase:** slow growth (small company, cautious hiring),
 - Middle phase:** rapid growth (expansion, mass hiring),
 - Later phase:** slowing down (reaching capacity or market saturation).

2. There Is a Natural Upper Limit

- Companies cannot hire employees indefinitely – there is a **maximum capacity**.
- Logistic regression naturally models this saturation, unlike linear models which assume infinite growth.

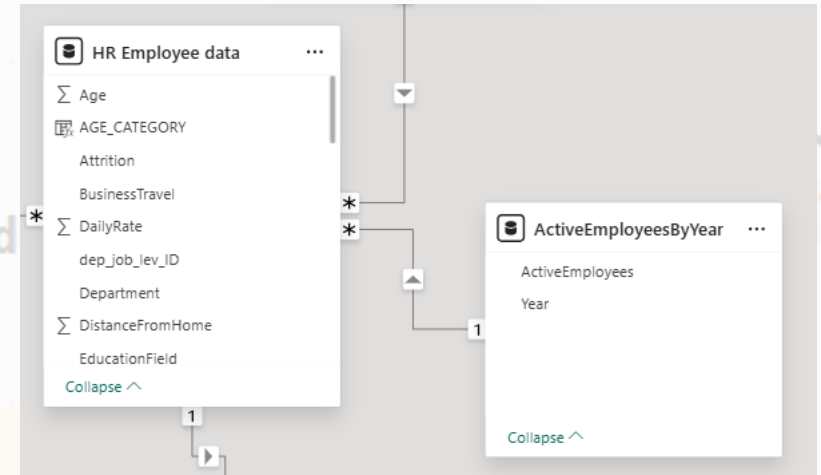
3. Ideal for Time Series with Saturation

- When tracking employee numbers over years, the trend often shows initial growth followed by stabilization or slight decline – a **classic logistic growth scenario**.

DAX formulas for creating the graph

```
NUMBER_OF_ACTIVE_EMPLOYEES =
CALCULATE(
    COUNT('HR Employee data'[EmployeeID]),
    ISBLANK('HR Employee data'[LeavingYear])
)
```

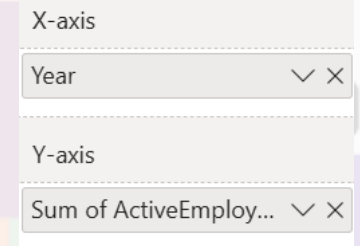
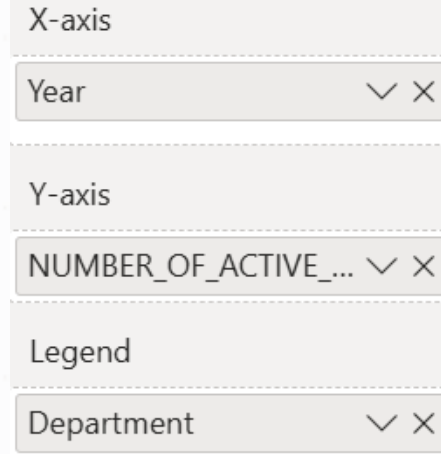
```
ActiveEmployeesByYear =
VAR YearsList = SELECTCOLUMNS(
    GENERATESERIES(
        MIN('HR Employee data'[JoiningYear]),
        MAX('HR Employee data'[LeavingYear]),
        1
    ),
    "Year", [Value]
)
RETURN
    ADDCOLUMNS(
        YearsList,
        "ActiveEmployees",
        CALCULATE(
            COUNTROWS('HR Employee data'),
            FILTER(
                'HR Employee data',
                'HR Employee data'[JoiningYear] <= [Year] &&
                (ISBLANK('HR Employee data'[LeavingYear]) || 'HR Employee
data'[LeavingYear] >= [Year])
            )
        )
    )
```



NUMBER_OF_ACTIVE_EMPLOYEES

3500
3000
2500
2000
1500
1000
500
0

Department ● Corporate Functions ● Delivery ● HR ● Marketing ● Prod



2008 2010 2012 2014 2016 2018 2020 2022

JoiningYear